

Stercoral Colitis: Review of Imaging Features and Complications

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Stercoral colitis is an inflammatory reaction secondary to fecal impaction and almost always occurs in the setting of chronic constipation. Luminal distention caused by dense and dehydrated stool compresses the vascular supply of the distal colon, resulting in bowel ischemia and ulceration. Stercoral colitis primarily affects elderly patients, but it can be seen in any patient with decreased bowel motility, with risk factors including neurodegenerative disorders, chronic medical diseases, malignancy, immobility, and the use of narcotic or anticholinergic medications. Patients most often present with abdominal pain and tenderness. However, the presentation is often nonspecific and can include nongastrointestinal symptoms. Due to the common presence of comorbidities, a thorough history and physical examination findings may be difficult to obtain. Imaging, especially CT, plays a vital role in the diagnosis of stercoral colitis, demonstrating hallmark features such as fecal impaction and a large colorectal stool burden. Mural thickening (>3 mm) and other CT signs of inflammation aid in diagnosis, although findings including perirectal fat stranding can be nonspecific. Signs of perforation, including mural discontinuity, extraluminal air, and extraluminal stool collections, can also be identified. Other potential complications include obstruction, bleeding, fistulas, and urinary tract involvement secondary to mass effect. The overlap of findings between stercoral colitis and other colonic diseases, particularly diverticulitis and malignancy, can sometimes make diagnosis challenging. Identification of fecal impaction and associated inflammatory changes helps in distinguishing stercoral colitis from other pathologic conditions. Prompt diagnosis of stercoral colitis and its complications allows appropriate management, which can range from preventive measures to emergent surgical treatment.



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Introduction

Stercoral colitis is an inflammatory reaction secondary to fecal impaction. Impacted stool can compress the vascular supply of the colon and rectum, causing ischemia and mucosal ulceration. It typically involves the rectosigmoid colon, although it can occasionally occur elsewhere in the colon (1–6). Stercoral colitis is associated with chronic constipation and

fecal impaction particularly among the elderly, in patients who receive opioids and other medications causing constipation, and in those with comorbidities including neurodegenerative disorders and chronic medical diseases.

Diagnosis can often be challenging due to nonspecific clinical presentations. The most common symptoms include nonspecific abdominal pain and/or distention and

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TEACHING POINTS

- The short segment of the colon between the branches of the inferior mesenteric and superior rectal arteries, known as the Sudeck point, marks the junction of the rectum and sigmoid colon. This region is a watershed territory with a relatively reduced blood supply and is susceptible to ischemia.
- The pathophysiology of stercoral colitis is primarily driven by bowel wall ischemia caused by luminal stool impaction.
- CT is the primary imaging modality to aid in diagnosing stercoral colitis and its potential complications.
- Identification of fecal impaction is essential for diagnosis. Inflammatory changes such as wall thickening and fat stranding typically are associated with the area affected by fecal impaction.
- Some of the imaging findings associated with worsened prognosis include long-segment (>40 cm) involvement of the colon and the presence of free fluid, extraluminal air, and extraluminal stool.

constipation (3–5). History and physical examination are often limited or unreliable due to the presence of neurologic and psychologic conditions, along with other comorbidities.

As a result, imaging often plays an important role in the diagnosis of stercoral colitis and its potential complications. However, there is a relative paucity of reported cases of stercoral colitis in the imaging literature and a lack of established diagnostic criteria (7–9). This has likely led to an underestimation of the true prevalence of stercoral colitis due to diagnostic challenges and its overlap of signs and symptoms with other conditions, including diverticulitis, infectious colitis, and colonic malignancies (5,7,8). This article highlights the key features of the disorder, with an emphasis on CT findings.

This topic is gaining relevance due to the rising incidence of stercoral colitis, likely due to an increase in the elderly population, the number of patients with comorbidities, and the tendency for stercoral colitis to be diagnosed as incidental to the primary reason for presentation. Moreover, the rising demand for imaging studies further underscores the relevance of this topic for radiologists.

Pathophysiology

The responsibility of the colon is to absorb water from waste to produce stool. Consequently, as fecal matter traverses the colon, it progressively becomes more inspissated and solid, collecting in the rectosigmoid region due to its narrower luminal caliber compared with that of the proximal colon (8,9). The short segment of the colon between the branches of the inferior mesenteric and superior rectal arteries, known as the Sudeck point, marks the junction of the rectum and sigmoid colon. This region is a watershed territory with a relatively reduced blood supply and is susceptible to ischemia (10). The most common anatomic locations for stercoral colitis to oc-

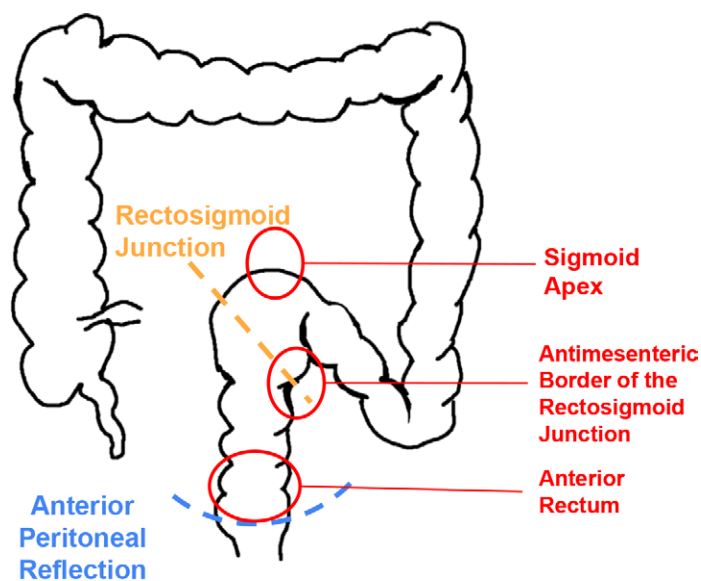


Figure 1. Illustration shows the most common anatomic sites affected by stercoral colitis.



Figure 2. Uncomplicated stercoral colitis in a 101-year-old woman who presented to the emergency department with lower abdominal pain. Axial contrast-enhanced CT image shows mild-to-moderate distention of the rectosigmoid junction with high-attenuation inspissated stool, perirectal edema (yellow arrow), and moderate presacral edema (white arrow). The relatively decreased blood supply in this region makes it prone to ischemia and stercoral ulceration.

cur include the apex of the sigmoid colon, the antimesenteric border of the rectosigmoid junction, and the anterior rectum (Figs 1, 2) (3,10,11). These sites are at increased risk due to the higher intraluminal pressures caused by decreased stool water content, smaller luminal diameter, and poorer perfusion (3). Approximately 77% of perforations associated with stercoral colitis occur in the rectosigmoid region, with the antimesenteric wall being the most at-risk site (8,9,11). Stercoral colitis uncommonly occurs in other portions of the colon, particularly with long-standing constipation that has reached the proximal colon (Fig 3A). In one study, Zacharias et al (6) reported a very small number of cases (1.9%) at a single medical institution that did not involve the sigmoid colon or

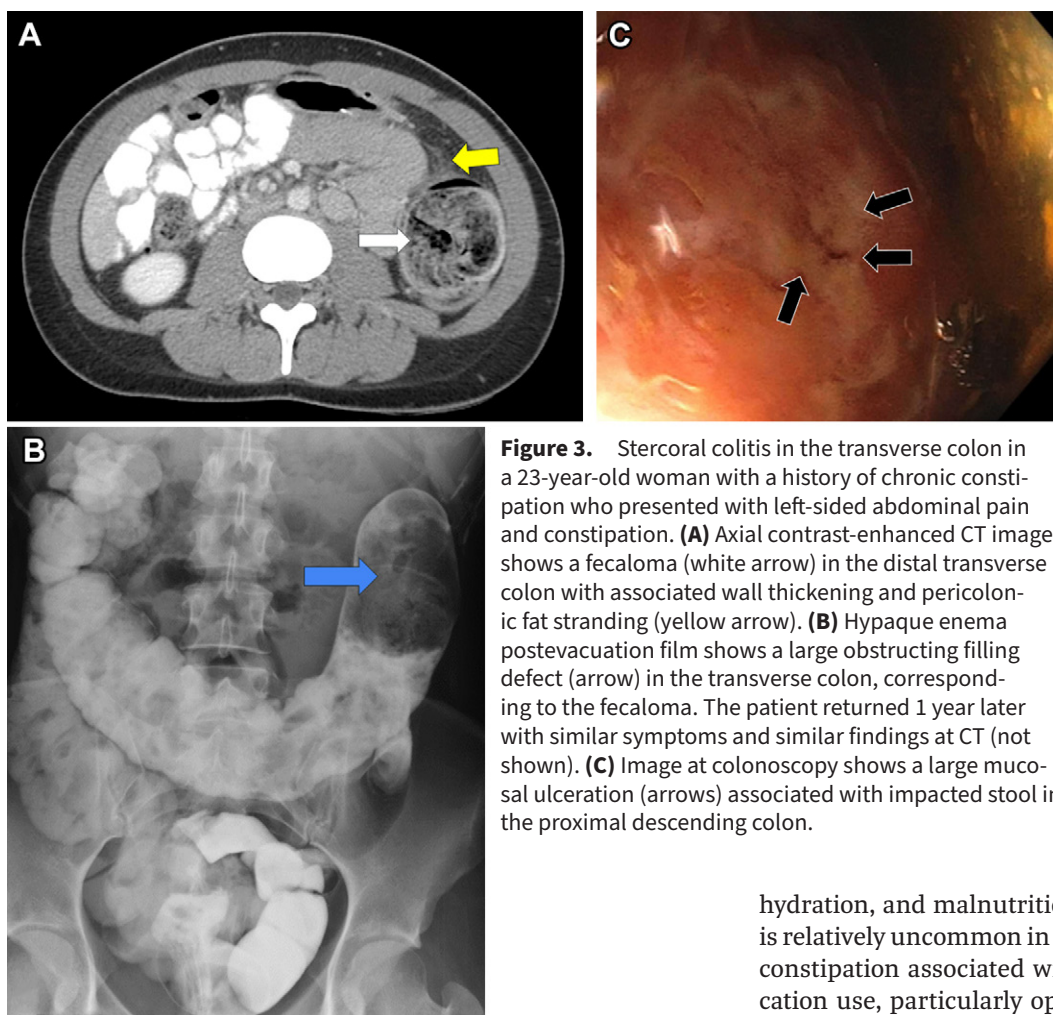


Figure 3. Stercoral colitis in the transverse colon in a 23-year-old woman with a history of chronic constipation who presented with left-sided abdominal pain and constipation. **(A)** Axial contrast-enhanced CT image shows a fecaloma (white arrow) in the distal transverse colon with associated wall thickening and pericolic fat stranding (yellow arrow). **(B)** Hypaque enema postevacuation film shows a large obstructing filling defect (arrow) in the transverse colon, corresponding to the fecaloma. The patient returned 1 year later with similar symptoms and similar findings at CT (not shown). **(C)** Image at colonoscopy shows a large mucosal ulceration (arrows) associated with impacted stool in the proximal descending colon.

rectum. Prior colonic surgery is a risk factor for the rare cases of stercoral colitis involving the proximal colon (6).

The pathophysiology of stercoral colitis is primarily driven by bowel wall ischemia caused by luminal stool impaction. Impacted fecal material in the colon causes compression of the colonic vasculature due to increased pressure on the wall (11). This results in ischemic injury and can lead to necrosis, ulceration (Fig 3C), and colonic wall perforation in a stepwise fashion (9,10,12). At pathologic examination, stercoral ulcers are described as irregular and “geographic,” matching the contour of the impacted stool (9). Histologic examination reveals ischemic pressure necrosis with thinning of the colonic wall and an inflammatory reaction, with transmural necrosis seen at perforation sites (8,9). The perforations are round or ovoid and typically occur at the center of the ulcers, where wall thinning is most pronounced (9).

Epidemiology

Stercoral colitis most commonly occurs in patients with chronic constipation, most often in the elderly, with a reported median age of 60–70 years (1,6,9,12,13). In addition to age, end-stage renal disease is a risk factor for constipation, possibly due to associated medications, fluid restriction, and decreased physical activity (14). Dietary issues can also be a contributing factor, particularly low-fiber diets, inadequate

hydration, and malnutrition (10,13,15). While stercoral colitis is relatively uncommon in younger patients, it can result from constipation associated with hospitalization and pain medication use, particularly opioids (16). Psychiatric medications including tricyclic antidepressants and antipsychotic medications are also linked to stercoral colitis, possibly due to their anticholinergic properties (9,17,18).

Comorbidities such as diabetes, hypertension, malignancy, immobility, neurologic disorders, metabolic abnormalities, and recent surgery may increase the risk for stercoral colitis through decreased physical mobility and gastrointestinal motility (1,4,5,7,9,15). Although the majority of stercoral colitis cases occur in elderly patients, cases in pediatric patients must not be overlooked. The rare pediatric cases of stercoral colitis may be due to chronic constipation, autism spectrum disorder, congenital anorectal malformations, and mental health disorders (19).

Clinical Presentation

Patients with stercoral colitis often present with nonspecific signs and symptoms. Abdominal pain and/or distention are the most common chief patient concerns. However, these symptoms are reported to be absent in over 60% of affected patients (13). Approximately one-third of patients do not have abdominal tenderness at physical examination (13). Signs of peritonitis may arise if perforation occurs, along with leukocytosis and elevated lactate levels in cases of ischemia (3,5,7,10). However, a recent study shows that many patients with stercoral colitis have a normal white blood cell count and lactate levels, so normal values do not necessarily exclude ischemia (13). About a quarter of patients present with nongastrointestinal

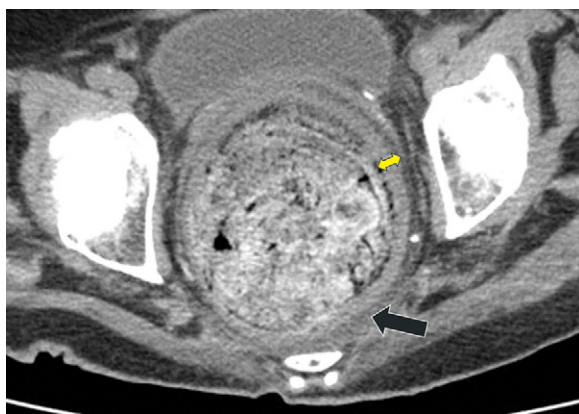


Figure 4. Uncomplicated stercoral colitis in a 75-year-old woman with Parkinson disease and dementia who presented to the emergency department from a nursing home with lethargy, fever, and leukocytosis. Further history was unable to be elicited from the patient. Axial noncontrast CT image shows moderate distention of the rectum with high-attenuation inspissated stool, wall thickening greater than 3 mm (double-headed yellow arrow), and perirectal fat stranding and presacral edema (black arrow).

issues, including altered mental status, generalized weakness, and urinary symptoms (13,20). Due to the nonspecific presentation and limitations of history taking and physical examination in this patient population, imaging plays a crucial role in the diagnosis of stercoral colitis.

Imaging Findings

Stercoral colitis is characterized by a large conglomerate stool burden, which is often visible as rectal and/or colonic distention on plain radiographs. However, this finding is nonspecific, and radiographs alone are insufficient to confirm the diagnosis. CT is the primary imaging modality to aid in diagnosing stercoral colitis and its potential complications. On CT images, stool typically has a “soap bubble” appearance. In stercoral colitis, inspissated stool forms hard masses known as fecalomas. These fecalomas are more dense compared with normal stool and often contain calcifications or have a laminated appearance (21). The impacted stool collects in the affected colon and rectum, resulting in the characteristic fecal loading observed in stercoral colitis (Fig S1). Distention of the rectal diameter beyond 6–6.5 cm is generally considered abnormal, but this should be compared with that of the proximal portions of the colon (5,22).

While fecal impaction and large stool burden are prerequisite CT findings, the diagnosis of stercoral colitis can be made only in the presence of mural abnormalities. Bowel wall thickening (>3 mm) at the site of stool impaction is an indicator of inflammation (12). The degree of bowel wall thickening may vary depending on the amount of distention, often appearing more subtle in cases of substantial colonic or rectal distention. Other signs of inflammation include perirectal fat stranding and presacral edema (Figs 2, 4), although these signs may be nonspecific because fluid and edema tend to accumulate in this dependent space. These findings should be used in conjunction with the presence of fecal impaction to help establish the diagnosis of stercoral colitis at CT.



Figure 5. Stercoral colitis involving an unusually long segment of the sigmoid colon in a 57-year-old woman with left lower quadrant abdominal pain, nausea, and anorexia. Axial (A) and coronal (B) CT images obtained with oral contrast agent show mild distention and a large fecal load throughout the visualized colon, with high-attenuation inspissated stool in the sigmoid. There is fat stranding and fluid surrounding the entire length of the sigmoid (arrows) and descending colon (not shown). At colonoscopy, a sigmoid stricture was found, likely accounting for the stool-filled colon. There was a 5-cm rock-hard stool ball in the sigmoid that was unable to be removed or fragmented by water lavage.

Aside from mural thickening, there are other CT abnormalities reported in patients with stercoral colitis. The “dense mucosa” sign, characterized by a high-attenuation colonic wall due to intramural hemorrhage, is best appreciated on noncontrast images and has been reported to be a sign of ischemic bowel (12,23) (Fig S2). Another sign is discontinuous mural enhancement that is manifested by low attenuation in the colonic wall secondary to hypoperfusion and may portend impending bowel perforation (5,12,24). Mucosal sloughing, visualized as a narrow irregular layer of bowel wall protruding into the colon, is thought to indicate the progression of bowel ischemia to infarction (4,24). Long-segment involvement of the colon (>40 cm) and the presence of free fluid have also been linked to increased mortality (5,24) (Fig 5).

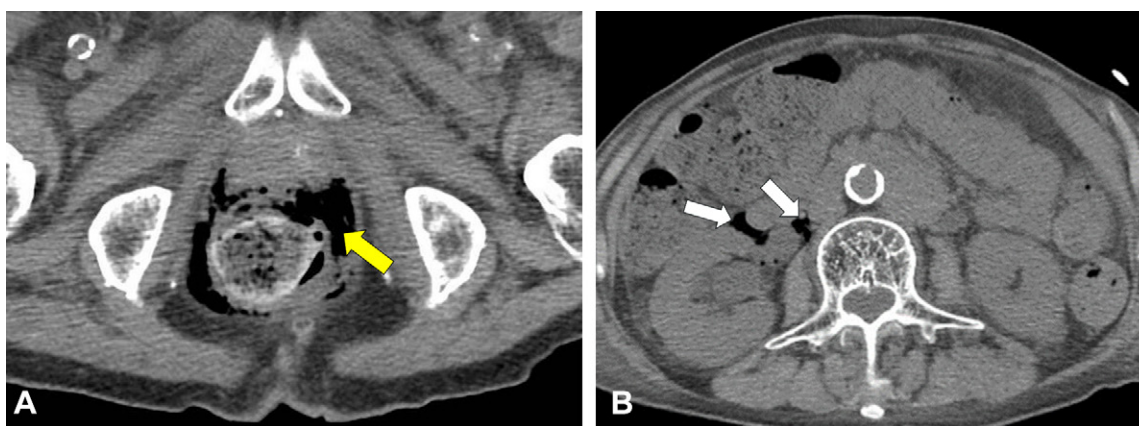


Figure 6. Stercoral perforation with extraperitoneal free air in a 91-year-old man who presented with failure to thrive. Axial noncontrast CT images show a rectal perforation with extensive perirectal free air (arrow in **A**) and extension of free air to the right upper retroperitoneum (arrows in **B**).

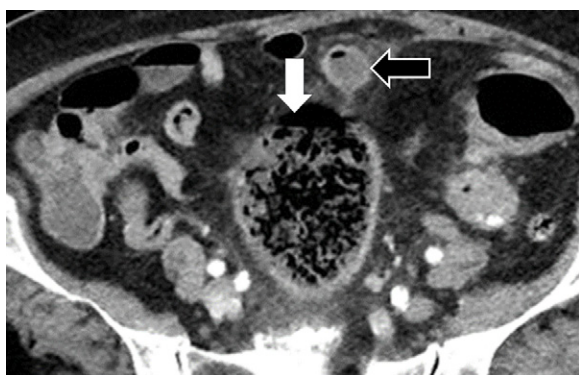


Figure 7. Perforated stercoral colitis in an 80-year-old woman who presented from a nursing home with abdominal pain and constipation. Initial CT image (not shown) showed evidence of stercoral colitis, which was managed medically. Follow-up axial contrast-enhanced CT image obtained 5 days later due to increasing leukocytosis shows a small fluid collection with a peripherally enhancing wall (black arrow) forming anterior to the rectosigmoid junction. There is a large defect in the anterior rectal wall (white arrow), which is the site of perforation.



Figure 8. Perforated stercoral colitis with extraluminal stool collection in an 87-year-old woman with a prior history of stercoral colitis who presented with abdominal pain and distention. Coronal contrast-enhanced CT image shows several pockets of extraluminal gas and fluid, as well as mural discontinuities involving the high rectum (white arrows). Extraluminal stool is present in the right pelvis adjacent to the rectum (black arrow).

Potential Complications

Extramural findings are associated with complications and are usually manifestations of ischemia leading to perforation. Stercoral perforation is associated with a mortality rate of 32%–59% due to extraluminal spillage of feces and the often rapid onset of feculent peritonitis (9). CT is particularly useful in the identification of direct sites of transmural injury or secondary signs of perforation, including mural discontinuity, extraluminal air, and perirectal fluid collection or abscess. (Figs 6–8). Extraluminal gas is easily identifiable on noncontrast CT images, but mural discontinuity may not be evident. Intravenous contrast agents can aid in the detection of sites of mural discontinuity, although sometimes this feature is discernible on noncontrast CT images (Fig 9). Ascites is a nonspecific finding. However, it is concerning in the absence of an obvious reason, including cardiac, liver, or kidney disease, or if it is a new finding in a patient with persistent symptoms or laboratory abnormalities (5). Pneumatosis can be seen as

a feature of stool ulcerating into the mucosa (4,11,13) (Fig 10). Extraluminal stool has the appearance of a mottled gas and low-attenuation soft-tissue collection without a surrounding wall and has been characterized as a “dirty mass” in previous reports. A dirty mass is often adjacent to a mural discontinuity, and this finding is highly specific for colonic perforation. In the authors’ experience and based on the limited available literature, this finding is concerning for the imminent development of life-threatening sepsis (8,24,25) (Fig 11). This finding can be difficult to identify, particularly when there is a redundant sigmoid colon filled with high-attenuation inspissated stool. It is important for radiologists to carefully trace the course of the redundant sigmoid to identify a dirty mass that diverges from the normal course of the colon, as this finding is worrisome for extraluminal stool (Fig 12).



Figure 9. Stercoral perforation in a 91-year-old woman who presented with hypoxia. Axial noncontrast CT image shows a large rectal stool burden with extraluminal gas and stool (black arrows) and two areas of mural discontinuity consistent with perforation (white arrows).



Figure 10. Perforated stercoral colitis in a 92-year-old woman with a history of dementia and bipolar disorder who was brought to the emergency department with fever and constipation. Axial contrast-enhanced CT image shows a small focus of gas along the left wall of the rectum (white arrow) and an extraluminal gas and fluid-containing collection along the right wall (black arrow).

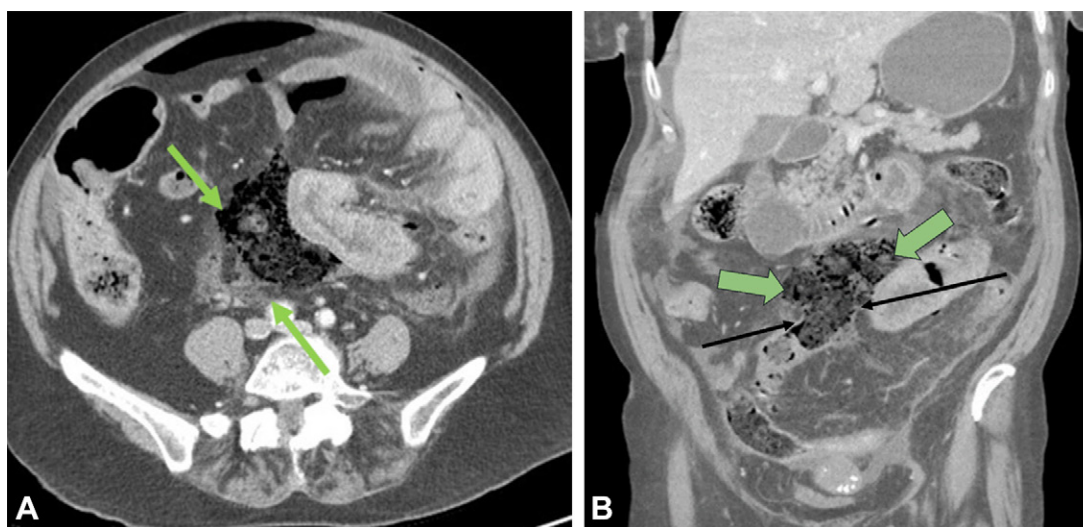


Figure 11. Feculent peritonitis from perforated stercoral colitis in a 71-year-old woman who presented with a 1-day history of severe abdominal pain, hypotension, and lactic acidosis. **(A)** Axial contrast-enhanced CT image shows a large mottled extraluminal collection without a surrounding wall, indicating extraluminal feces rather than an abscess (arrows). **(B)** Coronal CT reformation shows that the extraluminal stool (green arrows) is adjacent to a large mural defect at the sigmoid colon (black arrows).

Mural discontinuities and extramural gas are virtually diagnostic of perforation. Extraluminal air can collect in the intraperitoneal and/or extraperitoneal spaces, depending on the involved segment of the colon. In the authors' experience and based on the limited literature, stercoral rectal perforations are usually walled off and localized, which in theory may allow a better prognosis than an intraperitoneal perforation, which often leads to the rapid onset of frank sepsis (Fig S3) (7–9,26). However, the largest single-institution study that identified patients with stercoral colitis at CT showed that approximately 95% of stercoral perforation involved the sigmoid

colon, with only 5% of cases involving the rectum alone (6). This increased risk of perforation in the sigmoid colon is possibly related to the decreased luminal capacity and distensibility compared with that of the rectum (6). There are limited data regarding the incidence of perforation involving the more proximal colon; recent review articles have indicated an incidence of 11.1%–19.2% (27,28). The cecum, which has a significantly higher distensibility than the sigmoid colon, was implicated in less than 5.2% of cases (27,28).

While intravenous contrast agents can be helpful in identifying mural defects, the defects may be difficult to see when

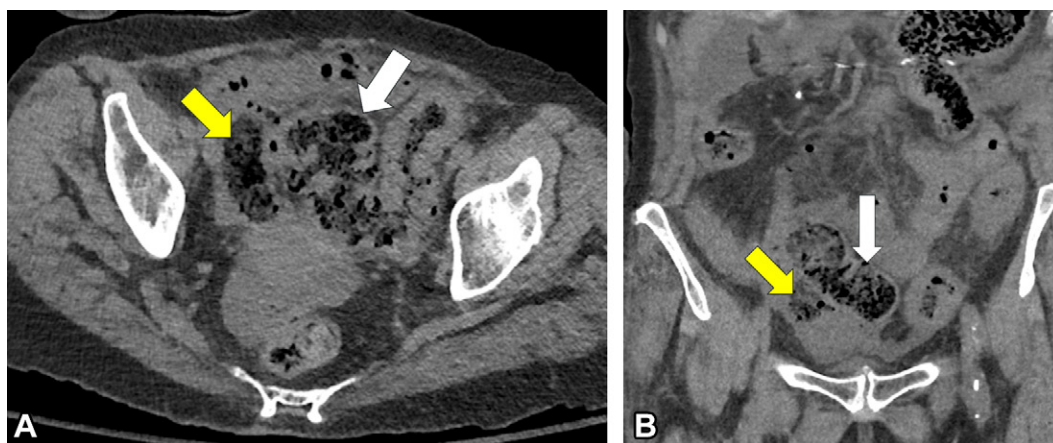


Figure 12. Perforated stercoral colitis with extraluminal stool in an 87-year-old woman with a history of bipolar disorder who presented with syncope. Axial (**A**) and coronal (**B**) noncontrast CT images show stool-filled loops of redundant sigmoid colon (white arrow). A “dirty mass” (yellow arrow) adjacent to the sigmoid colon is consistent with an extraluminal stool ball.

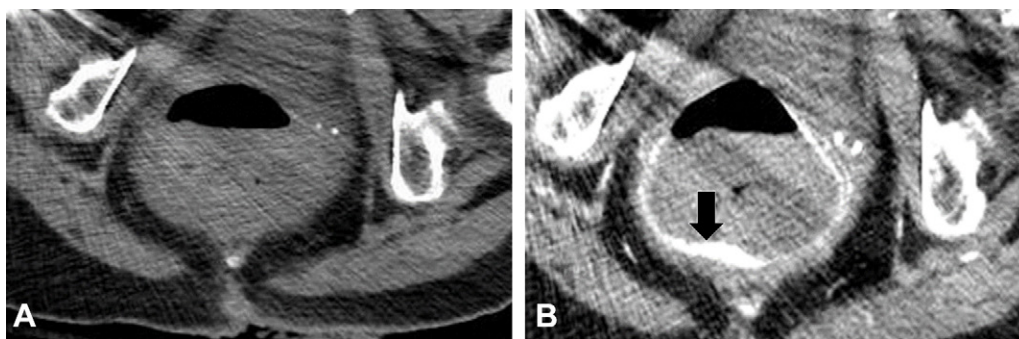


Figure 13. Acute gastrointestinal hemorrhage secondary to a stercoral ulcer in a 79-year-old woman with chronic constipation who presented with hematochezia. Diverticular bleeding was suspected, but the bleeding was found to be localized to the rectum. Axial noncontrast CT (**A**) and arterial phase (**B**) CT images of the rectum are shown. The arterial phase image (**B**) shows intravenous contrast agent extravasation pooling along the posterior wall of the lower rectum (arrow in **B**). A nonperforated stercoral ulcer was suspected to be the source of hemorrhage and was confirmed at colonoscopy.

small. In these instances, these small defects may be sealed by intraluminal stool (5). CT signs such as pelvic free fluid, peritoneal thickening, or peritoneal enhancement, reflective of peritonitis, may be present and are subtle indicators of perforation. With the emergence of dual-energy CT (DECT), this modality may have a possible role in the diagnosis of stercoral colitis. DECT has been shown to increase the sensitivity for detecting bowel ischemia (29). This feature of DECT may have a role in the diagnosis of stercoral colitis, as it is fundamentally caused by bowel ischemia from fecal impaction.

Stercoral colitis is a potential cause for complications that intrinsically involve the large bowel or affect other structures in the pelvis. Large bowel obstruction due to a sigmoid fecaloma is a relatively common finding and should be suspected when there is marked colonic dilatation proximal to the site of fecal impaction (30,31). Potential uncommon complications include lower gastrointestinal tract bleeding, fistulization, and urinary tract obstruction from mass effect on the bladder neck or ureters. Bleeding from nonperforated stercoral ulcers is a rare cause of hematochezia (32,33) (Fig 13). Although rare, fistulization to the adjacent structures from a focal perforation can also occur. In patients with constipation with bloody and feculent vaginal discharge, a rectovaginal fistula should be suspected (34) (Fig 14). Rectocutaneous fistulas have also been described in the literature (34). The large stool burden

associated with stercoral colitis can compress the bladder, resulting in urinary retention and hydronephrosis (2) (Fig 15).

The key imaging features of stercoral colitis are summarized in the Table. Many of these imaging findings are non-specific and can overlap with those of other pathologic bowel conditions. The key point is that identification of fecal impaction is essential for diagnosis. Inflammatory changes such as wall thickening and fat stranding typically are associated with the area affected by fecal impaction. Understanding the underlying pathologic cause of stercoral colitis will aid in distinguishing it from other forms of colitis.

Differential Diagnosis

Diagnosis of stercoral colitis at CT is often straightforward. However, findings from other colonic diseases may overlap. Some disease entities that may share similar findings include diverticulitis, ischemic colitis, large bowel obstruction, inflammatory bowel disease, pseudo-obstruction (Ogilvie disease), and colorectal malignancy.

Diverticulitis is an important differential consideration. There is an overlap in patient demographics, as it is most commonly seen in older patients with chronic constipation. In diverticulitis, wall thickening and fat stranding are usually centered around a single diverticulum, while the inflammation associated with stercoral colitis may be more diffuse, with

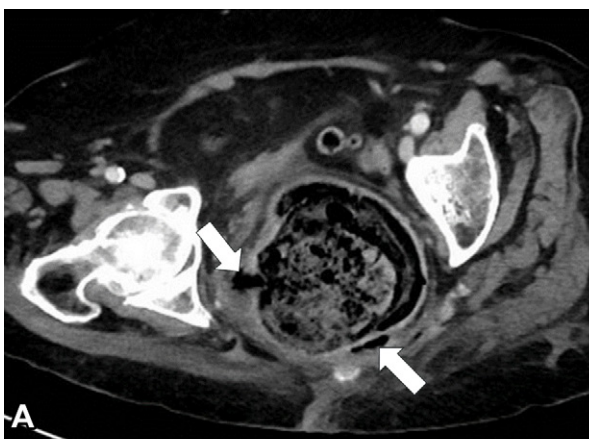


Figure 14. Stercoral colitis with rectovaginal fistula in a 93-year-old woman with a history of bipolar disorder, dementia, and hysterectomy who presented with abdominal pain and distention. **(A)** Axial contrast-enhanced CT image at presentation shows stercoral colitis with multiple rectal perforations (arrows). The patient was deemed a poor candidate for surgery and was therefore managed medically. **(B, C)** Axial **(B)** and sagittal **(C)** contrast-enhanced CT images obtained 1 month later when the patient presented with blood and feculent material in the vagina show a rectovaginal fistula (arrow) and distention of the rectum and vaginal cuff with hemorrhagic fluid.

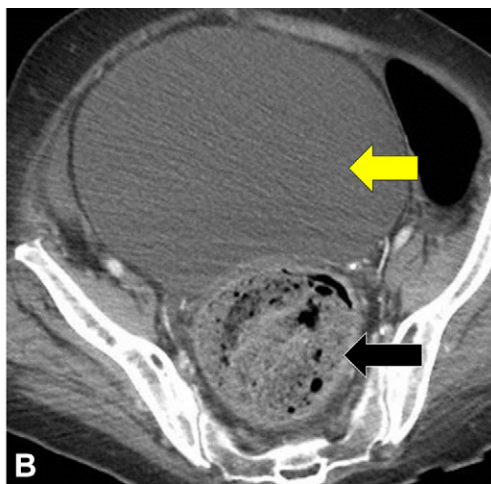
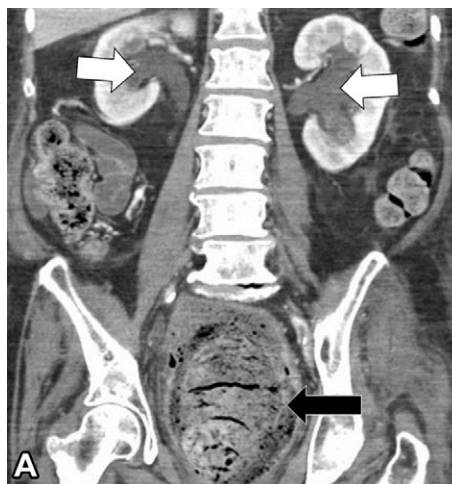
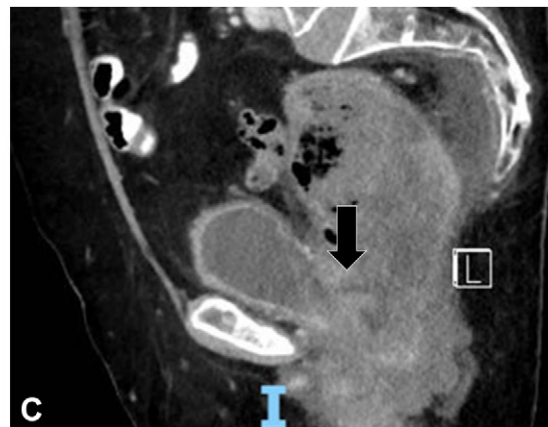
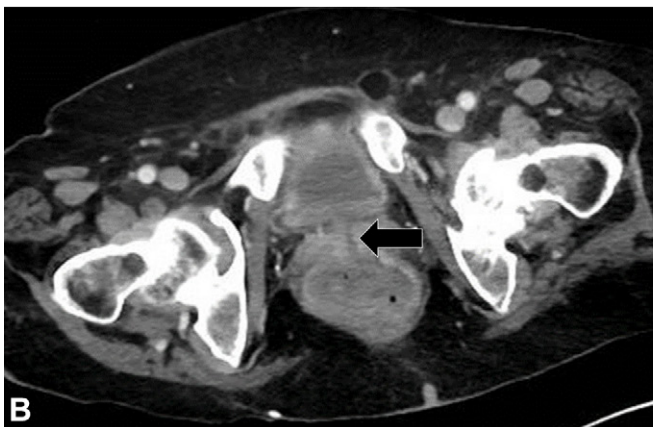


Figure 15. Stercoral colitis complicated by bladder outlet obstruction in a 75-year-old woman who presented with lethargy and leukocytosis. **(A)** Coronal contrast-enhanced CT image shows bilateral moderate hydronephrosis (white arrows) and a large rectal stool burden (black arrow). **(B)** Axial CT image of the pelvis shows a stool-filled distended rectum (black arrow) exerting mass effect on a markedly distended bladder (yellow arrow).

circumferential wall thickening and inflammatory changes involving the entire lumen (5) (Fig 16). Perirectal inflammatory changes favor stercoral colitis, as diverticula rarely form in the rectum. Perforated diverticulitis can resemble the extraluminal stool observed in perforated stercoral colitis, especially in cases in which a large diverticulum ruptures. However, the absence of colonic distention typically associated with fecal impaction favors diverticulitis as the likely underlying cause.

Furthermore, stercoral ulcers and perforation sites commonly occur at multiple locations, in contrast to diverticulitis (5).

Another diagnosis that has an overlapping patient demographic is an obstructive colorectal mass. There can be similarities in CT findings such as luminal distention and the buildup of stool proximal to the mass (25,30,35) (Fig 17). It is important to carefully evaluate for an obstructing mass when there is fecal impaction with a sudden transition to

Key Imaging Features of Stercoral Colitis

Imaging Finding	Description
Fecal impaction	Mass of inspissated and attenuating stool within the rectum and/or colon
Colonic distention	Increased caliber of bowel (<6 cm) at the site of fecal impaction
Colonic wall thickening	Increased wall thickness (<3 mm) at the site of fecal impaction
Pericolonic fat stranding	Increased attenuation of the surrounding fat
Mural discontinuity	Focal irregularity or defect in the normally smooth appearance of the colonic wall
Extraluminal air	Pneumatosis, portal venous gas, or free air within intraperitoneal and/or extraperitoneal spaces

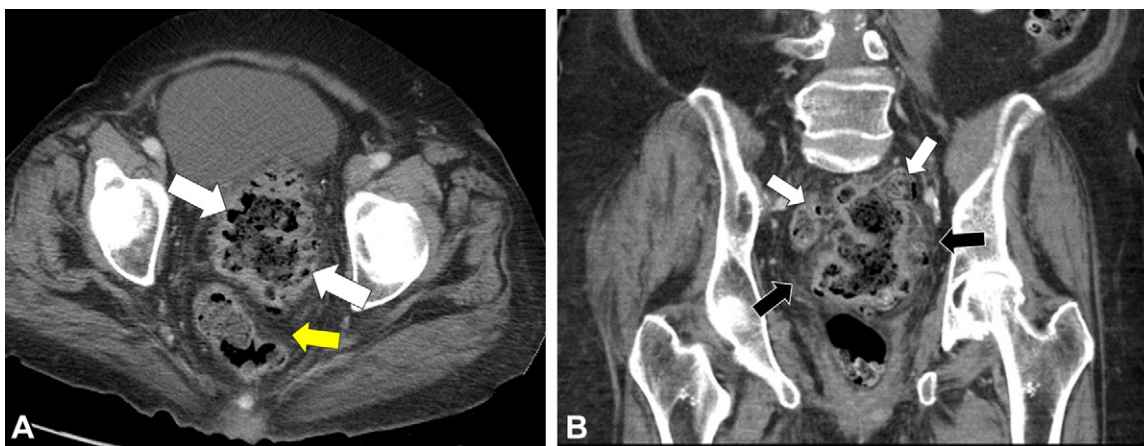


Figure 16. Acute sigmoid diverticulitis in an 82-year-old woman who presented with abdominal pain and nausea. Axial (**A**) and coronal (**B**) contrast-enhanced CT images show a stool-distended sigmoid colon with diverticula (white arrows) and pericolic fat stranding (black arrows in **B**). Note the lack of inflammation surrounding the rectum (yellow arrow in **A**). Stercoral colitis is a consideration for these findings. However, the presence of diverticulosis and sparing of the rectum favor acute diverticulitis.

underdistended colon. Imaging features that should increase the suspicion for underlying malignancy include pelvic lymphadenopathy, eccentric or irregular wall thickening, a “shouldering” appearance, and the presence of metastases (30,31,35,36). Although inflammatory conditions can also be associated with lymphadenopathy, the lymph nodes seen with malignancy are usually greater in size (>1 cm in short-axis diameter) and number (30,31,35,36).

A potential pitfall is the finding of presacral edema or fat stranding at CT, which may arise from a source other than the colon or rectum. Some common causes include inflammation from a decubitus ulcer; sacral insufficiency fractures; fibrosis related to rectal surgery or chemoradiation; and third spacing of fluid from hypoalbuminemia, renal disease, and congestive heart failure (37,38) (Fig 18). Radiation proctitis often results in severe circumferential wall thickening (39). The absence of a large fecaloma or rectosigmoid distention favors an alternative diagnosis. Some less common causes of presacral edema include a primary retroperitoneal or extraperitoneal tumor, metastatic disease, hemorrhage, osteomyelitis, and retroperitoneal fasciitis (37,38).

Management

Early recognition of stercoral colitis allows identification of at-risk patients and early intervention to reduce potential mor-

bidity and mortality. It is common for stercoral colitis to be diagnosed incidentally while a patient is seeking acute medical care (6). Therefore, appropriate and timely management can halt disease progression and prevent potential complications. Some of the imaging findings associated with a worsened prognosis include long-segment (>40 cm) involvement of the colon and the presence of free fluid, extraluminal air, and extraluminal stool (5,9,24).

Treatment is based on the severity of the disease. With uncomplicated stercoral colitis, a bowel regimen is used, which includes preventive measures such as a high-fiber diet and prophylactic use of laxatives and/or stool motility agents (3). It is advisable to discontinue anticholinergics or opioid medications when clinically feasible. Enemas and manual disimpaction are also recommended. Contrast agent enemas serve as an additional method for the diagnosis and treatment of stercoral colitis, particularly when large bowel obstruction is suspected (31). While their use is primarily documented in the older literature, they continue to be performed in an attempt to disimpact the fecaloma and mitigate the need for surgery (9) (Fig 3B).

Endoscopic disimpaction is often performed when the fecaloma is unreachable with digital disimpaction and enemas are not successful (10,11,13,19). Management can be challenging, with patients often requiring multiple procedures to adequately disimpact the stool. In the presence of

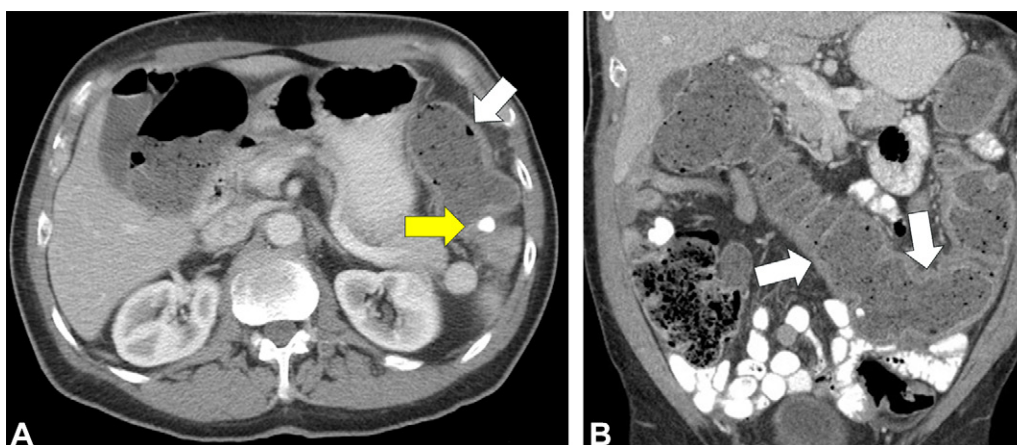


Figure 17. Colon cancer causing large bowel obstruction and stercoral colitis in a 72-year-old man who presented with lower abdominal pain. Axial (A) and coronal (B) CT images obtained with oral and intravenous contrast material show a large fecal load in the transverse colon with mural thickening (white arrows) and a subtle soft-tissue mass with shouldering at the splenic flexure (yellow arrow in A). This lesion was found to be a stage II adenocarcinoma.

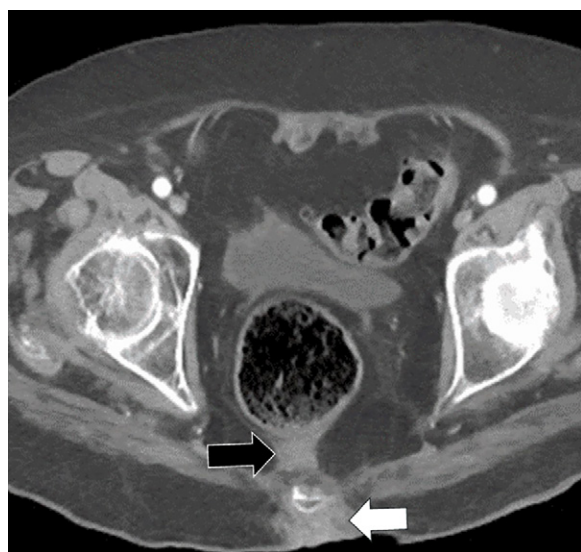


Figure 18. Sacral decubitus ulcer in a 65-year-old man who presented with fever. Axial contrast-enhanced CT image shows presacral edema and fat stranding (black arrow) and mild rectal distention with stool. However, there is no rectal wall thickening or perirectal fat stranding. An underlying decubitus ulcer is present (white arrow), explaining the presacral fat infiltration.

perforation, emergent surgical treatment with an end colectomy or a Hartmann procedure is usually necessary, along with evacuation of extraluminal fecal contents and peritoneal washout (40).

Conclusion

Stercoral colitis is essentially a form of ischemic colitis that is precipitated by chronic constipation. Although stercoral colitis is considered to be relatively rare in the literature, the authors believe this may be related to underdiagnosis due to nonspecific clinical presentations and overlap with other inflammatory bowel disorders. Stercoral colitis primarily affects the elderly, but the diagnosis should be considered in

any patient with decreased bowel motility and appropriate imaging findings. CT is the standard modality for diagnosis, allowing accurate identification of characteristic findings as well as potential complications. Awareness of this condition and its findings allows appropriate management, ranging from supportive measures to surgical treatment.

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References

- Gough AE, Donovan MN, Grotts J, Greaney GC. Perforated stercoral ulcer: a 10-year experience. *J Am Geriatr Soc* 2016;64(4):912–914.
- Hudson J, Malik A. A fatal faecaloma stercoral colitis: a rare complication of chronic constipation. *BMJ Case Rep* 2015;2015:bcr2015211732.
- Bae E, Tran J, Shah K. Stercoral colitis in the emergency department: a review of the literature. *Int J Emerg Med* 2024;17(1):3.
- Saksonov M, Bachar GN, Morgenstern S, et al. Stercoral colitis: a lethal disease-computed tomographic findings and clinical characteristic. *J Comput Assist Tomogr* 2014;38(5):721–726.
- Ünal E, Onur MR, Balci S, Görmez A, Akpınar E, Böge M. Stercoral colitis: diagnostic value of CT findings. *Diagn Interv Radiol* 2017;23(1):5–9.
- Zacharias NA, Lubner MG, Richards ES, Mao L, Pickhardt PJ. Stercoral colitis: CT imaging findings and clinical risk factors. *Abdom Radiol (NY)* 2023;48(10):3050–3062.
- Kwag SJ, Choi SK, Park JH, et al. A stercoral perforation of the rectum. *Ann Coloproctol* 2013;29(2):77–79.
- Maurer CA, Renzulli P, Mazzucchelli L, Egger B, Seiler CA, Büchler MW. Use of accurate diagnostic criteria may increase incidence of stercoral perforation of the colon. *Dis Colon Rectum* 2000;43(7):991–998.
- Serpell JW, Nicholls RJ. Stercoral perforation of the colon. *Br J Surg* 1990;77(12):1325–1329.
- Naseer M, Gandhi J, Chams N, Kulairi Z. Stercoral colitis complicated with ischemic colitis: a double-edge sword. *BMC Gastroenterol* 2017;17(1):129.
- Heffernan C, Pachter HL, Megibow AJ, Macari M. Stercoral colitis leading to fatal peritonitis: CT findings. *AJR Am J Roentgenol* 2005;184(4):1189–1193.

12. Wu CH, Wang LJ, Wong YC, et al. Necrotic stercoral colitis: importance of computed tomography findings. *World J Gastroenterol* 2011;17(3):379–384.
13. Keim AA, Campbell RL, Mullan AF, et al. Stercoral colitis in the emergency department: a retrospective review of presentation, management, and outcomes. *Ann Emerg Med* 2023;82(1):37–46.
14. Sumida K, Yamagata K, Kovessy CP. Constipation in CKD. *Kidney Int Rep* 2019;5(2):121–134.
15. Sacerdote M, Limback J, Zhu J. Fecal Impaction and Nonperforated Stercoral Colitis: Red Flags for Poor Outcomes. *Cureus* 2023;15(7):e41705.
16. Bolia R, Sherwani P, Garnaik DK. Stercoral colitis in an adolescent. *Clin Gastroenterol Hepatol* 2022;20(6):e1223.
17. Osman M, Fiorentino M, Gwiazdowski M, Gervasoni JE. Stercoral perforation in the setting of chronic antipsychotic use: a case report. *Cureus* 2023;15(1):e33631.
18. Canders CP, Shing R, Rouhani A. Stercoral colitis in two young psychiatric patients presenting with abdominal pain. *J Emerg Med* 2015;49(4):e99–e103.
19. Thomas MJ, Nanagiri A, Duvidovich S, Levy L, Bamji N. Mission disimpaction: endoscopic management of stercoral colitis in an adolescent. *JPGN Rep* 2023;4(2):e300.
20. Chowdhary R, Liu B, Husnain M. Stercoral colitis as a cause of altered mental status in a patient with cerebral palsy. *Cureus* 2022;14(12):e33040.
21. Fagelman D, Warhit JM, Reiter JD, Geiss AC. CT diagnosis of fecaloma. *J Comput Assist Tomogr* 1984;8(3):559–561.
22. Preston DM, Lennard-Jones JE, Thomas BM. Towards a radiologic definition of idiopathic megacolon. *Gastrointest Radiol* 1985;10(2):167–169.
23. Furukawa A, Kanasaki S, Kono N, et al. CT diagnosis of acute mesenteric ischemia from various causes. *AJR Am J Roentgenol* 2009;192(2):408–416.
24. Wu CH, Huang CC, Wang LJ, et al. Value of CT in the discrimination of fatal from non-fatal stercoral colitis. *Korean J Radiol* 2012;13(3):283–289.
25. Kothari K, Friedman B, Grimaldi GM, Hines JJ. Nontraumatic large bowel perforation: spectrum of etiologies and CT findings. *Abdom Radiol (NY)* 2017;42(11):2597–2608.
26. Tribelhorn E, Sadoun M, McClenathan J, Neal D. Extraperitoneal stercoral perforation of rectum. *ACS Case Rev Surg* 2018;1(5):10–14. https://www.facs.org/media/O1lhsypb/crs_v1n5_cr03.pdf.
27. Chakravartty S, Chang A, Nunoo-Mensah J. A systematic review of stercoral perforation. *Colorectal Dis* 2013;15(8):930–935.
28. Fernando S, Sarma DR. Rare but relevant: a systematic review of stercoral perforation. *Br J Hosp Med (Lond)* 2021;82(4):1–7.
29. Murray N, Darras KE, Walstra FE, Mohammed MF, McLaughlin PD, Nicolaou S. Dual-Energy CT in Evaluation of the Acute Abdomen. *Radiographics* 2019;39(1):264–286.
30. Verheyden C, Orliac C, Millet I, Taourel P. Large-bowel obstruction: CT findings, pitfalls, tips and tricks. *Eur J Radiol* 2020;130:109155 [Published correction appears in *Eur J Radiol* 2020;132:109284.]. <https://doi.org/10.1016/j.ejrad.2020.109155>.
31. Jaffe T, Thompson WM. Large-Bowel Obstruction in the Adult: Classic Radiographic and CT Findings, Etiology, and Mimics. *Radiology* 2015;275(3):651–663.
32. Hassranah S, Maharajh S, Solomon S, Naraynsingh V. Massive rectal hemorrhage from a stercoral ulcer. *Cureus* 2022;14(7):e26963.
33. Ahmad AI, Sebastian S, Govindu R, Ammar H. It is not diverticular bleeding! A case of stercoral ulcer. *Am J Med* 2021;134(4):e289–e290.
34. Maruyama T, Imamura F, Fukue M, Aoyagi H, Nozue M. Stercoral perforation of the rectum forming recto-vaginal and recto-cutaneous fistulae: a case report. *Nihon Gekakei Rengō Gakkai shi/Japanese College of Surgeons* 2003;28(6):1040–1043.
35. Moon JY, Hong SS, Hwang J, et al. Differentiation between stercoral perforation and colorectal cancer perforation. *Rev Assoc Med Bras* 2019;65(2):191–197.
36. Lips LMJ, Cremers PTJ, Pickhardt PJ, et al. Sigmoid cancer versus chronic diverticular disease: differentiating features at CT colonography. *Radiology* 2015;275(1):127–135.
37. Patel N, Maturen KE, Kaza RK, Gandikota G, Al-Hawary MM, Wasnik AP. Imaging of presacral masses--a multidisciplinary approach. *Br J Radiol* 2016;89(1061):20150698.
38. Chingko CM, Jahed A, Loreto MP, et al. Retroperitoneal fasciitis: spectrum of CT findings in the abdomen and pelvis. *Radiographics* 2015;35(4):1095–1107.
39. Guniganti P, Lewis S, Rosen A, Connolly S, Raptis C, Mellnick V. Imaging of acute anorectal conditions with CT and MRI. *Abdom Radiol (NY)* 2017;42(2):403–422.
40. Medellin Abueta A, Senejoa NJ, Pedraza Ciro M, et al. Laparoscopic Hartmann's reversal has better clinical outcomes compared to open surgery: An international multicenter cohort study involving 502 patients. *Health Sci Rep* 2022;5(5):e788.